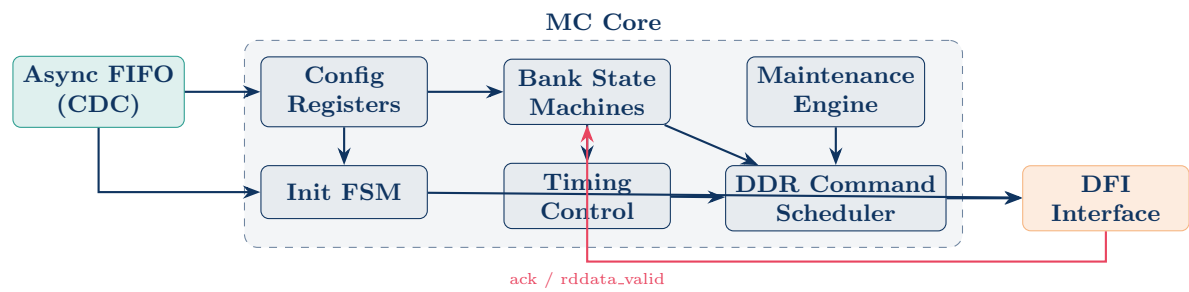


MC Core — Detailed Reference

Memory Controller Core · Sub-blocks · FSMs · Timing · Scheduling

DDR2/3/4/5 | LPDDR4/5 | DFI 5.2

1



2

Interface

APB/CSR slave → internal register file. Runtime-writable (live timing update gated by `ctrl_idle`).

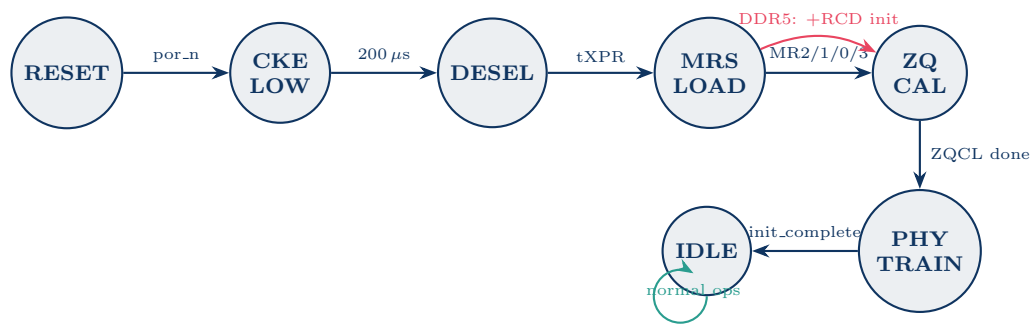
Key Register Groups

Group	Width	Contents
TIMING_*	8–16b each	tRCD, tCL, tRP, tRAS, tRFC, tRFC2, tFAW, tWR, tRTP, tCCD_L, tCCD_S, tWTR_L, tWTR_S, tRRD_L, tRRD_S
MR_SHADOW[0..6]	16b	Mirror of DRAM Mode Registers; scheduler reads for MRS cmd emission
DDR_VER	3b	Selects DDR2/3/4/5 or LPDDR4/5 → gates scheduler logic
PHY_OFFSET	8b	DFI <code>dfi_t_rddata_en</code> , <code>dfi_t_phy_wrlat</code> latency offsets
REFRESH_CFG	8b	tREFI override, PBR enable (DDR5), 2× refresh flag
ZQCAL_CFG	8b	tZQCS interval, ZQCL trigger
FREQ_RATIO	2b	1:1 / 1:2 / 1:4 → <code>dfi_freq_ratio</code> to PHY

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Function

Power-up sequencing, MRS loading, PHY training handshake. Emits DFI commands **directly**, bypassing scheduler.



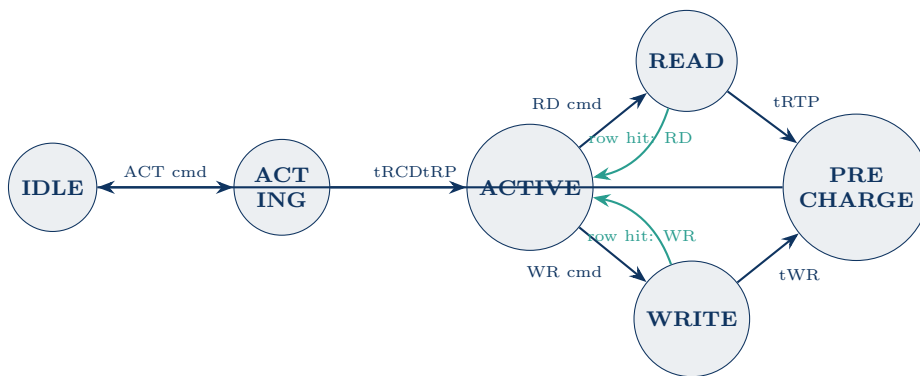
PHY Training Sequence (DFI handshake)

- Assert `dfi_init_start` → wait `dfi_init_complete`
- Write leveling: toggle `dfi_wrlvl_en`, sample `dfi_wrlvl_strobe`
- Gate training: `dfi_rdlvl_gate_en` / `dfi_rdlvl_resp`
- Read DQ training: `dfi_rdlvl_en` / `dfi_rdlvl_resp`
- VREF (DDR4/5 / LPDDR5): MRS-based sweep via Init FSM
- CA training (LPDDR5): `dfi_calvl_en`

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Instantiation

One FSM per bank × per rank. DDR4: 4 ranks × 16 banks = 64 FSM instances. DDR5 with bank groups: $4 \times 2 \text{ BG} \times 4 \text{ banks} = 32$.



Outputs to Scheduler

- `bank_available[N]`: bank not precharging / activating
- `open_row[N] [ROW_W]`: currently active row address
- `row_hit[N]`: incoming cmd address matches open row
- `bank_group_id[N]`: for `tCCD.L/S` differentiation

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Architecture

Per-bank countdown timers, decremented each MC clock. Output: `cmd_legal[N]` vector gating scheduler dispatch.

Timer	Source	Blocks
<code>tRCD</code>	ACT issue	RD/WR to this bank
<code>tRP</code>	PRE issue	ACT to this bank
<code>tRAS</code>	ACT issue	PRE to this bank (min open time)
<code>tFAW</code>	Sliding window	4-activation gate across all banks (same rank)
<code>tRFC</code>	REF issue	All cmds to rank
<code>tCCD.L</code>	CAS (same BG)	Next CAS same bank group
<code>tCCD.S</code>	CAS (diff BG)	Next CAS different bank group
<code>tWTR.L</code>	WR data end	RD cmd (same BG)
<code>tWTR.S</code>	WR data end	RD cmd (diff BG)
<code>tRTW</code>	RD cmd	WR cmd: $t_{CL} + t_{BL/2} + 2 - t_{CWL}$
<code>tRRD.L/S</code>	ACT	Next ACT same/diff BG
<code>tZQCS</code>	ZQCS issue	All cmds

tFAW Sliding Window

4-entry shift register of ACT timestamps. $\text{faw_ok} = (\text{now} - \text{act_ts}[3]) \geq t_{FAW}$

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6.1 Refresh Manager

- **Counter:** counts down t_{REFI} ($7.8\mu\text{s}$ DDR4, $3.9\mu\text{s}$ at $T > 85^\circ\text{C}$)
- **Credit accumulation:** up to 8 postponed REFs (JEDEC allows burst of 8 back-to-back)
- **Priority escalation:** if credits $\geq 8 \rightarrow$ assert `ref_urgent` $\rightarrow$ scheduler stalls all others
- **DDR5 PBR:** per-bank refresh — issues `REFpb`(bank) instead of `REFab`
- **LPDDR5:** `tRFCpb` vs `tRFCab` selection based on `REFRESH_CFG[PBR_EN]`

6.2 ZQ Calibration

- **ZQCL:** init only / forced via `ZQCAL_CFG[ZQCL_TRIG]`; stalls all cmds for `tZQoper`
- **ZQCS:** periodic short calibration every `tZQCS` interval; 1 cycle stall only
- **PHY-side:** some DFI implementations expose `dfi_calvl_periodic` for automatic PHY ZQ

6.3 Temperature Monitor

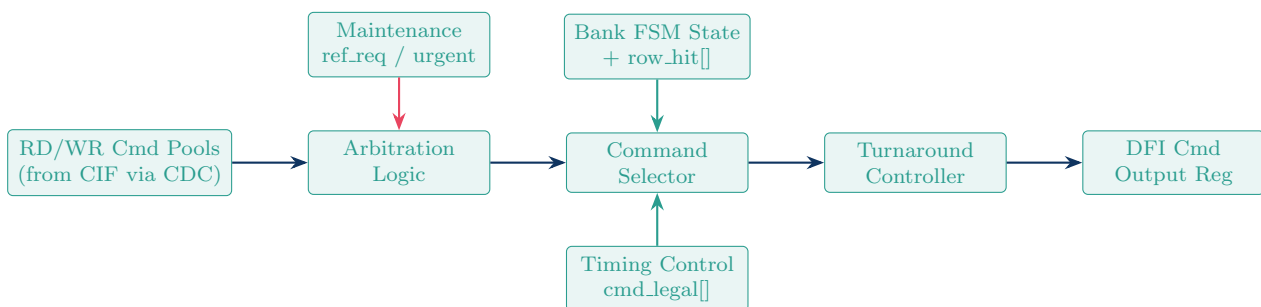
- Polls DRAM MR4 (DDR3/4) or MR18 (LPDDR5) via periodic MRR
- Thresholds: $T > 85^\circ\text{C} \rightarrow t_{REFI} \div 2$; $T > 95^\circ\text{C} \rightarrow$ derating applied to all timing params
- Output: `derating_en` $\rightarrow$ Timing Control adds derating offsets

6.4 Scrubber (ECC configs)

- Background read patrol: walks full address space at low priority
- On ECC error: triggers corrective write-back (SEC correctable)
- DED (double-bit) $\rightarrow$ raises `irq_ecc_fatal`

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Pipeline Diagram



7.1 Arbitration Logic

Priority order (highest $\rightarrow$ lowest):

1. Refresh urgent (`ref_urgent` asserted)
2. Refresh nominal (`ref_req`)
3. ZQ calibration
4. Read commands (latency-sensitive)
5. Write commands

6. Maintenance (scrubber, MRR)

Starvation prevention: Age counter per entry. If $\text{age} \geq \text{AGE_THR}$ → priority promoted to top of its class.

7.2 Command Selector

Selects best legal command from arbitrated pool:

- Checks `bank_available[N]` AND `cmd_legal[N]`
- **Open-page policy (default):** if `row_hit` → issue CAS directly; skip ACT
- **Closed-page:** force PRE after every access (streaming/random workloads)
- **Adaptive:** switch policy based on hit-rate counter threshold
- Bank-group scheduling: prefer `tCCD_S` path (different BG) for back-to-back CAS

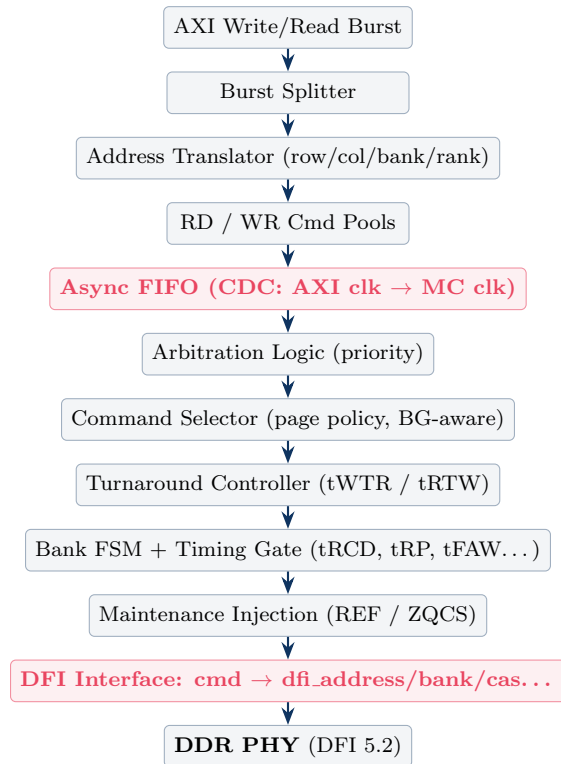
7.3 Read/Write Turnaround

Transition	Constraint	Expression
WR → RD (same BG)	<code>tWTR_L</code>	$t_{CWL} + t_{BL/2} + t_{WTR_L}$
WR → RD (diff BG)	<code>tWTR_S</code>	$t_{CWL} + t_{BL/2} + t_{WTR_S}$
RD → WR	<code>tRTW</code>	$t_{CL} + t_{BL/2} + 2 - t_{CWL}$

DFI pipeline offset applied: `dfi_t_rddata_en` cycles ahead of CAS for `dfi_rddata_en`; `dfi_t_phy_wrlat` for `dfi_wrdata_en`.

7.4 Output Format (per cycle)

Field	Width	Meaning
<code>cmd_type</code>	3b	ACT / RD / WR / PRE / REF / NOP / MRS / ZQCS
<code>rank</code>	2b	Target rank
<code>bank_group</code>	2b	DDR4/5 bank group (0 for DDR3)
<code>bank</code>	2b	Target bank
<code>row</code>	17b	Row address (ACT)
<code>col</code>	11b	Column address (CAS)
<code>ap</code>	1b	Auto-precharge flag



Reference: JEDEC JESD79-5B (DDR5), JESD209-5B (LPDDR5), MIPI DFI 5.20 spec. Timing values device-specific; consult datasheet for exact values.